

Research Statement – Max Schroeder

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My research studies how the supply and demand for skills shape labour market outcomes, inequality, and broader macroeconomic dynamics. A central theme of this agenda is understanding how human capital accumulation interacts with individual heterogeneity, particularly in wealth, to generate persistent differences in economic outcomes. Recent papers in this area analyse the changing distribution of graduate skills in the UK and examine how the joint distribution of wealth and human capital influences aggregate responses to economic shocks and policy interventions.

Methodologically, my work draws on quantitative economic models, which I estimate or calibrate using rich microdata to uncover latent mechanisms and conduct theoretically disciplined counterfactual analysis. This approach allows me to bridge detailed micro-level behaviour with aggregate and distributional outcomes in a transparent and policy-relevant way.

Alongside this core research agenda, I maintain a strong interest in what historical data can teach us about economic persistence. In this strand of my work, I study the long-run dynamics of infectious disease and the enduring effects of cultural persistence on housing markets in Northern Ireland, using historical variation to shed light on phenomena that are difficult to identify in modern datasets.

Higher Education and Skill Formation

My first research agenda studies education and skill formation, with a focus on how higher education shapes labour market outcomes and inequality. This work examines how changes in the supply of graduate skills interact with labour market demand to shape occupational sorting and wages.

My job market paper, [Skills, Tasks and Degrees](#), develops and estimates a quantitative model of occupational choice for UK university graduates. The model treats skills as a latent variable inferred from observed occupation choices and wages, allowing me to separately identify changes in the distribution of graduate skills, the returns to skill across occupations, and the role of non-pecuniary job attributes. Estimating the model using UK microdata from 2001–2019, I document a substantial decline in average graduate skill levels and show that this shift explains the majority of the observed reduction in professional employment among young graduates. Changes in returns to skill and job amenities further contribute to the reallocation of graduates toward routine occupations.

This paper speaks directly to the current policy debate surrounding the value of higher education in the UK and is currently under review at *Labour Economics*. Building on this work, I am developing a research agenda to study the effects of expanding participation initiatives in higher education, examining how widening access affects the composition of skills, labour market matching, and inequality.

Wealth and Human Capital Inequality in Macroeconomic Models

My second research agenda studies how the joint distribution of wealth and human capital shapes macroeconomic dynamics and the distributional effects of economic shocks and policy interventions. This work moves beyond models that treat human capital as exogenous, highlighting how heterogeneity in wealth feeds back into human capital accumulation and amplifies inequality over time.

In a chapter of my PhD thesis ([link](#)), I develop a heterogeneous-agent macroeconomic model with endogenous accumulation of both wealth and human capital. The model shows that the distribution of wealth plays a central role in shaping human capital investment decisions, leading to persistent inequality and asymmetric responses to aggregate shocks. In particular, households with limited wealth face tighter constraints on human capital accumulation, which slows adjustment and magnifies the long-run distributional consequences of adverse economic shocks.

Building on this framework, my ongoing work with Spyridon Lazarakis extends the model to include government debt and fiscal consolidation. This extension allows us to study the full costs and benefits of fiscal adjustment when households differ in both their wealth positions and human capital. The analysis shows that consolidation affects households heterogeneously across the wealth distribution and that alternative policy instruments—such as labour versus asset taxation—have markedly different distributional and welfare consequences once human capital accumulation is endogenous.

Together, this strand of my research provides a framework for analysing macroeconomic policy that explicitly accounts for endogenous human capital dynamics and distributional heterogeneity and complements my work on education by embedding individual skill accumulation in a general equilibrium setting.

Persistence and Learning from Historical Data

A related strand of my research uses historical data to study economic persistence and to identify mechanisms that are difficult to uncover using modern datasets alone. This work exploits archival sources to analyse how cultural and health-related factors shape economic outcomes over long time horizons.

In work published in *Social Science & Medicine* (joint with S. Lazarakis, R. Mancy, and K. Angelopoulos), we study the long-run dynamics of infectious disease using historical data on influenza outbreaks. The paper documents substantial persistence in mortality risk following major epidemic waves, showing that health shocks can generate elevated risks long after the acute phase has passed. These findings highlight the importance of accounting for slow-moving health dynamics when assessing the long-run economic and distributional consequences of pandemics and related policy responses.

In related work on housing markets in Northern Ireland (joint with R. Cho, H. Farag, C. Görtz, D. McGowan, and H. Nguyen), we examine how long-run cultural persistence influences contemporary economic outcomes. Using variation in historical religious composition, we show that areas with a history of intercultural mixing exhibit substantially higher property prices today, even after controlling for a rich set of housing and neighbourhood characteristics. The findings attracted significant public and policy interest, including coverage by major media outlets, and the paper is currently in preparation for submission to a leading economics journal ([working paper](#)).

Building on this work, I have developed a broader research agenda on the origins of long-run cultural persistence in Northern Ireland and have prepared a grant proposal to the British Academy to study the role of religious integration and local cultural institutions using archival sources.

Measuring Expectations and Constraints Using Text-Based Methods

More recently, I have become involved in a project that develops new empirical measures of financial constraints using textual analysis of corporate disclosures. In joint work with R. Cho, C. Görtz, and D. McGowan, we construct a novel firm-level measure of financial constraints from firms' 10-K filings that explicitly distinguishes between constraints that are currently binding and those expected to bind in the future.

The key contribution of this paper is to show that accounting for the timing of constraints reveals economically meaningful heterogeneity that standard measures overlook. We document systematic differences in balance sheet behaviour and transition dynamics across firms facing current versus expected constraints, and we use this distinction to provide the first direct empirical test of the theoretical mechanism underlying the cash flow sensitivity of cash. While we confirm that this sensitivity captures firms anticipating future constraints, we show that widely used empirical implementations conflate unconstrained and currently constrained firms, leading to a misinterpretation of this canonical measure. This work revisits a foundational result in corporate finance and is currently being prepared for submission ([working paper](#)).

More broadly, this work has motivated an ongoing interest in using modern text-based methods to construct economically interpretable measures of latent expectations across a range of economic settings.

Taken together, my research agenda combines quantitative economic modelling with rich micro, historical, and textual data to address policy-relevant questions about skills, human capital, and inequality. Looking ahead, this approach provides a flexible platform for studying a range of related topics, including the labour-market effects of technological change, the interaction between education policy and macroeconomic adjustment, and the role of institutions, expectations, and persistence in shaping long-run economic outcomes. By grounding these analyses in structurally interpretable models, my work aims to produce empirically disciplined insights that inform both academic research and policy design.